

Panel Data Econometrics

Chapter 5 – Some advanced issues *discussing what comes next*

Introduction

In this fifth chapter, we discuss advanced issues, which we may not have time to cover deeply in this class.

Part 1 – Testing for homogeneity

Part 2 – Dynamic panel data models

Testing for homogeneity

The pooled assumption

So far, in this course, we have considered that all individuals from the panel share a common Data Generating Process, except for the random error.

→ We assumed that $\beta_{ik} = \beta_k \forall i$

This is called the « pooled assumption ».

One can however argue that individuals may vary not only in the level of y , but also in the relationship between y and X .

Testing for homogeneity

The heterogeneous model

$$y_{it} = \beta_0 + \beta_{1i}x_{1i} + \beta_{2i}x_{2i} + \cdots + \beta_{Ki}x_{Ki} + \varepsilon_{it}$$

In such a model, there are $N \times K$ coefficients to estimate (against K if you assume homogeneity).

Note: If we assume no fixed effect, we would need $N \times (K - 1) + 1$

If you have data with few individuals, and many periods, this model can be well estimated. If $T < K$ this model cannot be estimated!

Testing for homogeneity

The heterogeneous model

$$y_{it} = \beta_0 + \beta_{1i}x_{1i} + \beta_{2i}x_{2i} + \cdots + \beta_{Ki}x_{Ki} + \varepsilon_{it}$$

In the end, this model is not much better than doing the regressions one by one on each individuals... This becomes time-series observations and we would reject the panel structure.

Assuming heterogeneity can however be interesting when we apply it to a subset of variables, and not to all of them.

$$ProdEvaluation_{it} = \beta_0 + \beta_1 Sponsored_{it} + \beta_{2i} Price_{it} + \varepsilon_{it}$$

Testing for homogeneity

The heterogeneous model

$$ProdEvaluation_{it} = \beta_0 + \beta_1 Sponsored_{it} + \beta_{2i} Price_{it} + \varepsilon_{it}$$

In this model, the effect of being a sponsored evaluator (compared to not being so) is captured by β_1 . We assume that it is the same effect for everyone, as we want a « general effect ».

The model takes into account the idea that reaction to the price of the product can vary from an evaluator to the other by including **N coefficients** related to price.

Testing for homogeneity

The heterogeneous model

$$ProdEvaluation_{it} = \beta_0 + \beta_1 Sponsored_{it} + \beta_{2i} Price_{it} + \varepsilon_{it}$$

For N=3, the model is :

$$\begin{aligned} ProdEvaluation_{it} \\ = \beta_0 + \beta_1 Sponsored_{it} + \beta_{21} Price_{1t} \\ + \beta_{22} Price_{2t} + \beta_{23} Price_{3t} + \varepsilon_{it} \end{aligned}$$

How would you do it using a statistical software ?

Testing for homogeneity

The heterogeneous model

$$Price_i = Price \times I_{ind=i}$$

ind.	t	Sponsor	Price	Price1	Price2	Price3
1	1	1	23	23	0	0
1	2	0	20	20	0	0
1	3	0	25	25	0	0
2	1	1	26	0	26	0
2	2	1	14	0	14	0
2	3	0	25	0	25	0
3	1	0	43	0	0	43

Testing for homogeneity

How to test which specification works best?

To test $H_0: \beta_i = \beta \forall i$ on L variables.

We use the Fisher's test :

$$F = \frac{RSSR - USSR}{USSR} \times \left[\frac{(N \times T - L \times N - (K - L))}{L(N - 1)} \right] \sim F_{(L(N-1), N \times T - L \times N - (K - L))}$$

With $RSSR = \sum_{i,t} e_{it}^2$ for the restrained model (ex: X includes price)

With $USSR = \sum_{i,t} e_{it}^2$ for the unconstrained model (ex: X includes price1, price2 and price3)

Dynamic panel data models

The bias of the OLS estimator : Nickell (1981)

Let us consider a model such as :

$$income_{it} = \beta_0 + \beta_1 income_{i\{t-1\}} + Coaching_{it} + \varepsilon_{it}$$

and let us assume that there are individual effects

$$\varepsilon_{it} = \eta_i + v_{it}$$

The OLS and the GLS estimator will be biased.

Indeed, $income_{i\{t-1\}}$ and ε_{it} are not independent : they both include the individual effect η_i !

Dynamic panel data models

The bias of the OLS estimator

How to solve this issue is still a hot debate !

- Anderson and Hsiao (1982) : Use the first-difference form to eliminate individual effects
- Arrelano and Bond (1991) : GMM and instrumental variable estimator.
- Ahn and Schmidt (1995), Blundell and Bond (1998), Hahn (1999)
- ...
- Hausman and Pinkovskiy (2017)
- Beck *et al.* (2017)

Dynamic panel data models

The bias of the OLS estimator

Packages on R to use these more advanced methods are available and quite well documented.

For instance, the pdynmc package ...

<https://cran.r-project.org/web/packages/pdynmc/vignettes/pdynmc-introLong.pdf>

... includes GMM methods for dynamic panel data model, including the recent iterated GMM estimator from Hansen and Lee (2019).